CURRICULUM VITAE

Personal information

Jing Hua Zhao
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Education

- PhD, Statistical Genetics (supervisor: Prof Pak Sham), King's College London, UK

Professional positions

2005.9-2018.7	Investigator scientist in Genetics, MRC Epidemiology Unit
2018.8-	Genetic Analyst / Senior Research Associate, Cardiovascular Epidemiology Unit, Department of Public Health and Primary Care, University of Cambridge

Research interests

My work is human health-related research which over years includes familial aggregation, segregation, linkage, candidate genes and genomewide association studies (GWASs). The most recent is proteogenomics within the SCALLOP consortium using the Olink and mass spectrometry (MS) panels measured for the INTERVAL samples. I have also led collaborative analysis to the SCALLOP-Seq(uence, both WES and NGS) consortium and Host Genetics Initiative.

I have promoted reproducible research through CRAN (https://cran.r-project.org), GitHub (https://github.com) and websites. I developed genetic analysis package (gap), protein quantitative trait tools (pQTLtools) and curated https://jinghuazhao.github.io/Computational-Statistics/, https://jinghuazhao.github.io/software-notes/ and https://jinghuazhao.github.io/Omics-analysis/. By closely following up developments in computational statistics, machine learning and artificial intelligence, I have made computing and omics analysis tools available from the University HPC, https://cambridge-ceu.github.io/csd3/systems/ceuadmin.html, whose components include AI with BitNet, Claude Code, GeminiCLI, Ollama, OpenClaw, llama.cpp, featuring AI for MS data with InstaNovo & DIA-NN, molecule optimization with DrugAssist, single-cell omics with Seurat, scp, scanpy, scvi-tools, scGPT, C2S-Scale, mtDNA analysis with MToolBox, fNUMT, haplogrep as well as long-read sequencing analysis with SVanalyzer, hap.py, sniffles, truvari. Explorations in population genetics include selscan, angsd, relate, clues2 and fastsimcoal2. These are coupled with extensive experiments on various models, architectures, platforms as with libraries such as TensorFlow, PyTorch, scikit-llm, LangChain.

Key publications

Zhao JH, et al. Mapping pQTLs of circulating inflammatory proteins identifies drivers of immune-mediated disease risk and novel therapeutic targets. *Nat Immunol* 2023, **24**(9):1540-1551, 10.1038/s41590-023-01588-w, https://www.nature.com/articles/s41590-023-01588-w.

COVID-19 Host Genetics Initiative. Mapping the human genetic architecture of COVID-19. *Nature* **600**:472–477 (2021); Pathak, G.A. et al. A first update on mapping the human genetic architecture of

COVID-19. *Nature* **608**:E1-E10 (2022); The Host Genetics Initiative. A second update on mapping the human genetic architecture of COVID-19. *Nature* **621**:E7–E26 (2023).

Zhao JH, Luan JA, Congdon P. Bayesian linear mixed model of polygenic effects. *J Stat Soft.* 2018, **85**(6):1-27. doi: 10.18637/jss.v085.i06

Zhao JH, Luan JA. Mixed modeling with whole genome data. *J Prob Stat*. 2012. doi: 10.1155/2012.485174.

Xue F, Li S, Luan J, Yuan Z, Luben RN, Khaw K-T, Wareham NJ, Loos RJF, **Zhao JH**. A latent variable partial least squares path modeling approach to regional association and polygenic effect with applications to a human obesity study. *PLoS ONE* 2012, **7**(2): e31927

Loos RJ, *et al*. Common variants near *MC4R* are associated with fat mass, weight and risk of obesity. *Nat Genet* 2008; **40**(6):768-75

Zhao JH. gap: genetic analysis package. *J Stat Soft* 2007, 2**3** (8):1-18. doi: 10.18637/jss.v023.i08.

Publications since 2018

188. Demenais F, et al. Multi-ancestry genome-wide association study identifies new asthma susceptibility loci that co-localize with immune cell enhancer histone marks. *Nat Genet* 2018; 50(1):42-53
189. Medina-Gomez C, et al. Life-course genome-wide association study meta-analysis of total body BMD and assessment of age-specific effects. *Am J Hum Genet* 2018; 102(1):88-102
190. Sung YJ, et al. A large-scale multi-ancestry genome-wide study accounting for smoking behavior identifies multiple significant loci for blood pressure. *Am J Hum Genet* 2018, 102(3):375-400. doi: 10.1016/j.ajhg.2018.01.015.
191. Turcot V, et al. (2018). Protein-altering variants associated with body mass index implicate pathways that control energy intake and expenditure underpinning obesity. *Nat Genet* 50:26-41
192. **Zhao JH**, Luan JA, Congdon P. Bayesian linear mixed model of polygenic effects. *J Stat Soft* 2018, 85(6):1-27. doi: 10.18637/jss.v085.i06
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194. Lee JJ, et al. Gene discovery and polygenic prediction from a genome-wide association study of educational attainment in 1.1 million individuals. *Nat Genet* 2018, 50:1112–1121, https://doi.org/10.1038/s41588-018-0147-3.
195. Ligthart S, et al. Genome analyses of >200,000 individuals identify 58 loci for chronic inflammation and highlight pathways that link inflammation and complex disorders. *Am J Hum Genet* 2018, 103(5):691-706. doi: 10.1016/j.ajhg.2018.09.009.
196. Evangelou E, et al. Genetic analysis of over 1 million people identifies 535 new loci for blood pressure traits. *Nat Genet* 2018, 50(10):1412-1425. doi: 10.1038/s41588-018-0205-x.
197. Merino J, et al. Genome-wide meta-analysis of macronutrient intake of 91,114 European ancestry participants from the cohorts for heart and aging research in genomic epidemiology consortium. *Mol Psychiatr* 2018 Jul 9. doi: 10.1038/s41380-018-0079-4.
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203. Shrine N, et al. New genetic signals for lung function highlight pathways and chronic obstructive pulmonary disease associations across multiple ancestries, *Nat Genet 2019,* https://doi.org/10.1038/s41588-018-0321-7.
204. Zhao J, et al. Meta-analysis of genome-wide association studies provides insights into genetic control of tomato flavor. *Nat Comm* 2019, 10, Article number: 1534 (2019), https://www.nature.com/articles/s41467-019-09462-w
205. Sung YJ, et al. A multi-ancestry genome-wide study incorporatinggene–smoking interactions identifies multiple new locifor pulse pressure and mean arterial pressure. *Hum Mol Genet* 2019, doi: 10.1093/hmg/ddz070
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207. Bentley A.R., *et al.* Multi-ancestry genome-wide gene–smoking interaction study of 387,272 individuals identifies new loci associated with serum lipids. *Nat Genet* **51,** 636–648 (2019). https://doi.org/10.1038/s41588-019-0378-y
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211. Cuellar-Partide G, et al. Genome-wide association study identifies 48 common genetic variants associated with handedness. *Nat Hum Behaviour* 2021, 5: 59–70.
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214. COVID-19 Host Genetics Initiative. Mapping the human genetic architecture of COVID-19. *Nature* 2021.
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216. Graham SE, et al. The power of genetically diverse individuals in genome-wide association studies of blood lipid levels. *Nature* 2021, **600**: 675–679, DOI:10.1038/s41586-021-04064-3
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